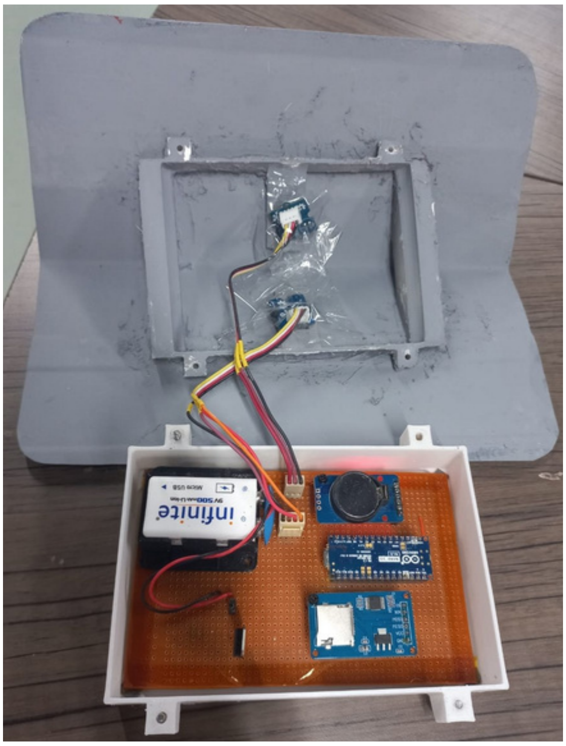
Trail test-1
NON-MECHANICAL RAIN GAUGE

METHOD: By using Tap

Date :25-10-2022
Time Duration: 5 minutes data

HARDWARE SETUP



Front view

OUTPUT DATA

```
Time= : 14:8:22, Date (D/M/Y) =  25/10/2022
sound   5
loudness        124
Time= : 14:9:22, Date (D/M/Y) =  25/10/2022
sound   148
loudness        121
Time= : 14:10:22, Date (D/M/Y) =         25/10/2022
sound   52
loudness        127
Time= : 14:11:22, Date (D/M/Y) =         25/10/2022
sound   587
loudness        41
Time= : 14:12:22, Date (D/M/Y) =         25/10/2022
sound   555
loudness        113
Time= : 14:13:23, Date (D/M/Y) =         25/10/2022
sound   21
loudness        278
Time= : 14:14:23, Date (D/M/Y) =         25/10/2022
sound   164
loudness        272
Time= : 14:15:23, Date (D/M/Y) =         25/10/2022
sound   11
loudness        185
Time= : 14:16:23, Date (D/M/Y) =         25/10/2022
sound   57
loudness        127
Time= : 14:17:23, Date (D/M/Y) =         25/10/2022
sound   72
loudness        120
```

TAP OFF

```
Time= : 14:19:23, Date (D/M/Y) =         25/10/2022
sound   207
loudness        238
Time= : 14:20:23, Date (D/M/Y) =         25/10/2022
sound   24
loudness        212
Time= : 14:21:24, Date (D/M/Y) =         25/10/2022
sound   177
loudness        316
Time= : 14:22:24, Date (D/M/Y) =         25/10/2022
sound   61
loudness        261
Time= : 14:23:24, Date (D/M/Y) =         25/10/2022
sound   52
loudness        105
Time= : 14:24:24, Date (D/M/Y) =         25/10/2022
sound   67
loudness        142
Time= : 14:25:24, Date (D/M/Y) =         25/10/2022
sound   59
loudness        103
```

TAP ON

TRIAL TEST 2
NON MECHANICAL RAIN GAUGE

METHOD: By using Mist Irrigation Nozzles

Date: 27/10/2022
TIME DURATION: 45 minutes data

HARDWARE SETUP:

NOZZLES

COLLECTION SETUP

SETUP FOR SIMULATING LIGHT RAIN AND HEAVY RAIN

DATA PRE-PROCESSING

Moderate Rain

Time	Sound	Loudness
12:07:07	659	260
12:08:07	20	271
12:09:08	620	273
12:10:08	486	238
12:11:08	521	258
12:12:08	200	254
12:13:09	491	91

Set up 1
Settings used for creating this condition:

Tap was almost closed.

Heavy Rain

Time	Sound	Loudness
12:15:09	447	277
12:16:09	280	255
12:17:10	308	224
12:18:10	472	260
12:19:10	383	287
12:20:10	12	139

Setup 2
Settings used for creating this condition:

Tap was more closed than the previous set

Light Rain

Time	Sound	Loudness
12:21:11	39	141
12:22:11	26	131
12:23:11	137	141
12:24:12	254	165
12:25:12	130	136
12:26:12	19	102

Setup 3
Settings used for creating this condition:

*Tap was more open than **setup 1***

Modified Nozzle (i)

Time	Sound	Loudness
12:32:14	90	172
12:33:14	447	275
12:34:14	665	240
12:35:15	563	251

Setup 4
Settings used for creating this condition:

The Mist Nozzle was cut.
Tap was partially open.

Modified Nozzle (ii)

Time	Sound	Loudness
12:37:15	848	383
12:38:16	676	373
12:39:16	377	378
12:40:16	362	328
12:41:17	469	372
12:42:17	56	120

Setup 5
Settings used for creating this condition:

The Mist Nozzle was cut.
Tap was more open than previous setup.

Fan jet Nozzle

Time	Sound	Loudness
12:43:17	79	139
12:44:18	615	338
12:45:18	79	126

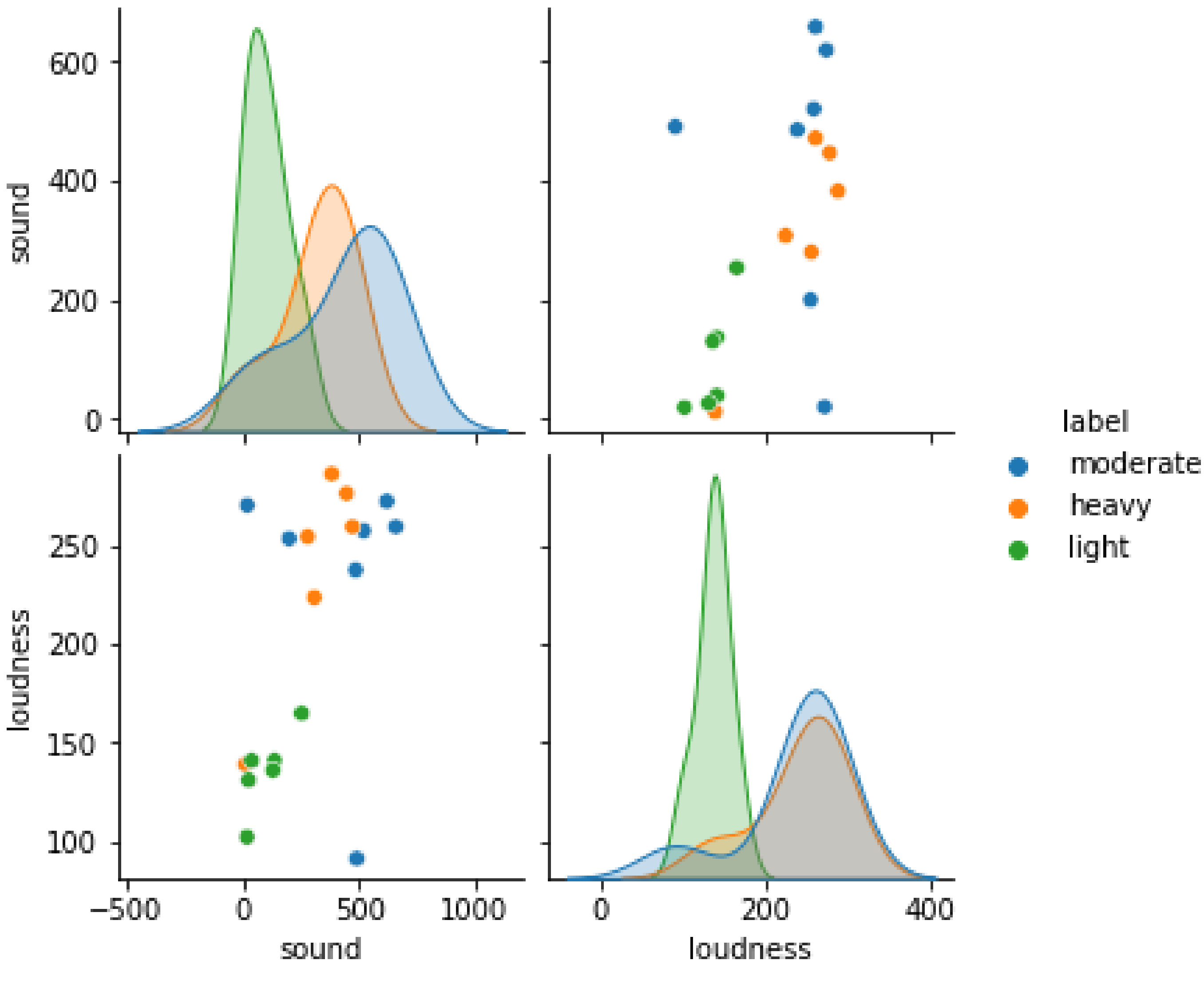
Setup 6
Settings used for creating this condition:

The Fan Jet Nozzle was used
Tap was same as previous setup.

Labelled data

Sound	Loudness	Label
659	260	moderate
20	271	moderate
620	273	moderate
486	238	moderate
521	258	moderate
200	254	moderate
491	91	moderate
447	277	heavy
280	255	heavy
308	224	heavy
472	260	heavy
383	287	heavy
12	139	heavy
39	141	light
26	131	light
137	141	light
254	165	light
130	136	light
19	102	light

CLASSIFIED LABELLED DATA



POWER CONSUMPTION

Date:31/10/2022

Theoretical Analysis

By using Component datasheet,
The maximum and minimum values of power consumption were calculated.

1.RTC Module

Operating Voltage= 3.63 V - 5.5 V
Current Requirement= 200 μA - 300 μA
Power = V*I
Minimum Power= 3.63*200*10^-6
= **0.000726 W**

Maximum Power= 5.5*300*10^-6
= 0.00165 W

2.Micro SD Adapter

Operating Voltage= 4.5 V - 5.5 V
Current Requirement= 0.2 mA - 200 mA
Minimum Power=4.5*0.2*10^-3
= **0.0009 W**

Maximum Power= 5.5*200*10^-3
= 1.1W

3.Sound Sensor

Operating Voltage= 5 V
Current Requirement= 4 mA ~ 5 mA
Minimum Power=5*4*10^-3
= **0.02W**

Maximum Power= 5*5*10^-3
= 0.025W

4.Loudness Sensor

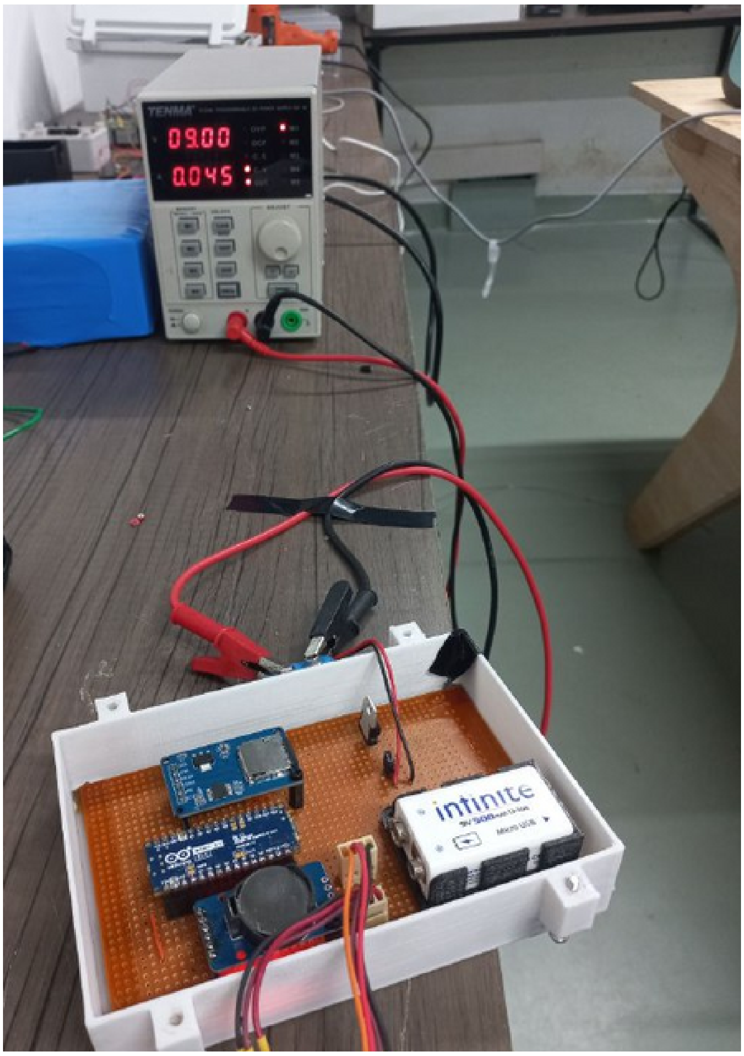
Operating Voltage=3.5 V ~ 10 V

Power Rating Minimum= 0.0009+0.000726+0.020
= 0.021626 Watts

Power Rating Maximum= 1.1+0.00165+0.025
=1.12665 Watts

Average Power Rating is 0.5 Watts

Practical Analysis



Practical Demonstration

Voltage = 9V
Current = 0.045 A
Power = V*I
= 9*0.045
= 0.405W

Overall Power Consumption of the non mechanical rain gauge is 0.405 Watts

Estimation Of Battery Life

- For a 500mAh Li-ion battery,

$$\textbf{Battery Life} = \frac{\textbf{Capacity(mAh)}}{\textbf{Circuit Current(mA)}}$$

$$= \frac{0.5\ \text{Ah}}{0.045\ \text{A}}$$

=11.11hour

=11 hour 6 min

- For a 800mAh Li-ion battery,

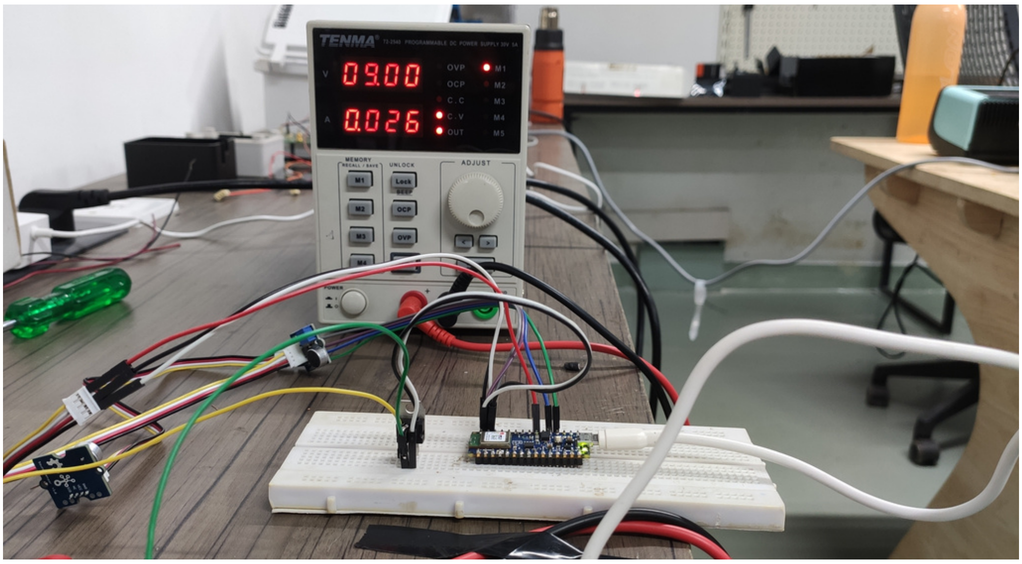
$$\textbf{Battery Life} = \frac{\textbf{Capacity(mAh)}}{\textbf{Circuit Current(mA)}}$$

$$= \frac{0.8\ \text{Ah}}{0.045\ \text{A}}$$

=17.77hour

=17 hour 46min

Overall power consumption of Arduino Nano BLE ,Sound sensor,Loudness sensor,7805 voltage regulator



Voltage = 9V
Current_{min} = 0.024 A
Power_{min} = V*I
= 9*0.024
= 0.216W

Voltage = 9V
Current_{max} = 0.026 A
Power_{max} = V*I
= 9*0.026
= 0.234W

Average current=0.025A
Power =V*I
=9*0.025
=0.225 W

Overall power consumption for Data Collection setup (Arduino Nano BLE,Sound sensor,Loudness sensor,7805 voltage regulator)= 0.225 Watts

- For a 500mAh Li-ion battery,

$$\textbf{Battery Life} = \frac{\textbf{Capacity(mAh)}}{\textbf{Circuit Current(mA)}}$$

$$= \frac{0.5\ \text{Ah}}{0.025\ \text{A}}$$

$$= \underline{\textbf{20 hour}}$$

- For a 800mAh Li-ion battery,

$$\textbf{Battery Life} = \frac{\textbf{Capacity(mAh)}}{\textbf{Circuit Current(mA)}}$$

$$= \frac{0.8\ \text{Ah}}{0.025\ \text{A}}$$

$$= \underline{\textbf{32 hour}}$$

POWER CONSUMPTION

Date:31/10/2022

Theoretical Analysis

By using Component datasheet,
The maximum and minimum values of power consumption were calculated.

1.RTC Module

Operating Voltage= 3.63 V - 5.5 V
Current Requirement= 200 µA - 300 µA
Power = V*I
Minimum Power= 3.63*200*10^-6
= **0.000726 W**

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2.Micro SD Adapter

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Current Requirement= 0.2 mA - 200 mA
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3.Sound Sensor

Operating Voltage= 5 V
Current Requirement= 4 mA ~ 5 mA
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Maximum Power= 5*5*10^-3
= 0.025W

4.Loudness Sensor

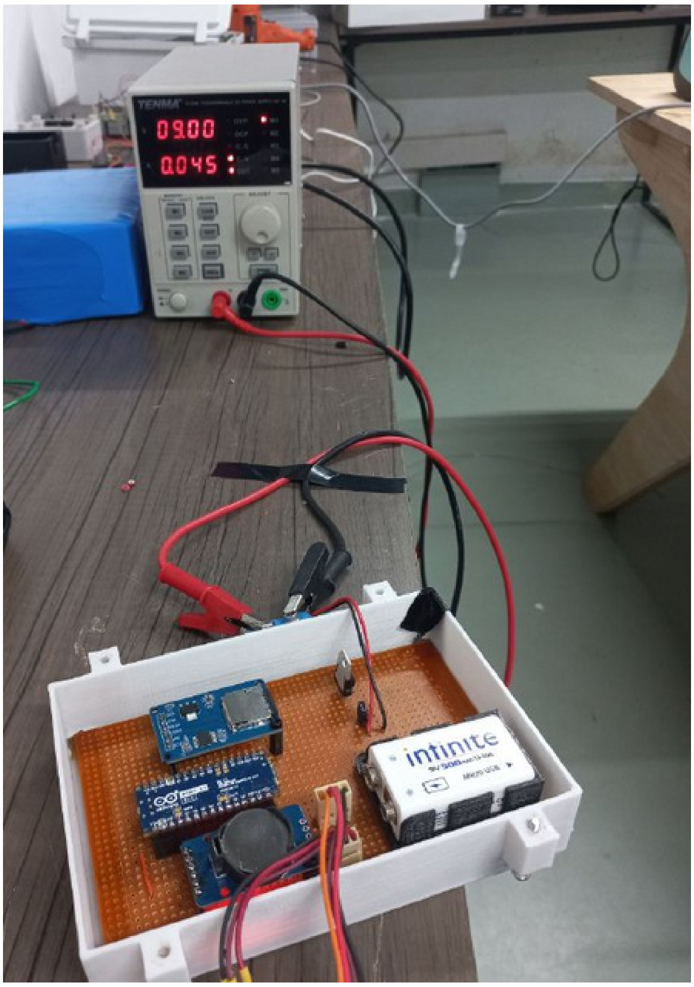
Operating Voltage=3.5 V ~ 10 V

Power Rating Minimum= 0.0009+0.000726+0.020
= 0.021626 Watts

Power Rating Maximum= 1.1+0.00165+0.025
=1.12665 Watts

Average Power Rating is 0.5 Watts

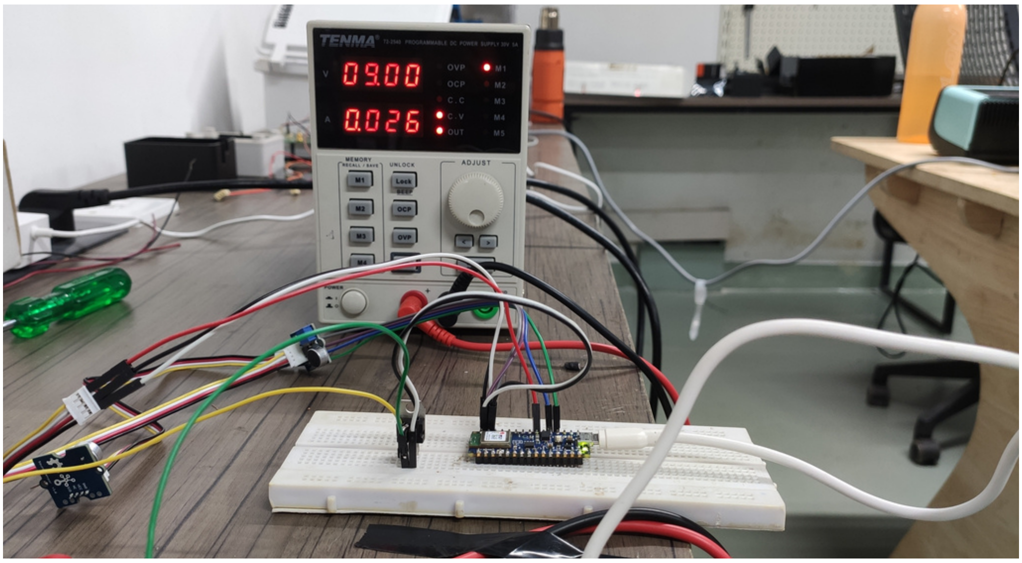
Practical Analysis



Setup

Voltage = 9V
Current = 0.045 A
Power = V*I
= 9*0.045
= 0.405W

Overall power consumption of Arduino Nano BLE ,Sound sensor,Loudness sensor,7805 voltage regulator



Voltage = 9V
Current_{min} = 0.024 A
Power_{min} = V*I
= 9*0.024
= 0.216W

Voltage = 9V
Current_{max} = 0.026 A
Power_{max} = V*I
= 9*0.026
= 0.234W

Average current=0.025A
Power =V*I
=9*0.025
=0.225 W

Overall power consumption for Data Collection setup (Arduino Nano BLE,Sound sensor,Loudness sensor,7805 voltage regulator)= 0.225 Watts

Overall power consumption for Data Collection setup (Arduino Nano BLE,Sound sensor,Loudness sensor,7805 voltage regulator,microSD adapter, RTC)= 0.405 Watts

Capacity

- **for 500 mAh=11 Hr 6 min**
- **for 800 mAh=17 Hr 46 min**

Overall power consumption for Edge device (Arduino Nano BLE sense,Sound sensor,Loudness sensor,7805 voltage regulator)= 0.225 Watts

Capacity

- **for 500 mAh=20 Hr**
- **for 800 mAh=32 Hr**

In a practical analysis, an 800mAh battery has a 7-hour capacity.